Machine Learning Models for Objective Assessment of Vascular Anastomoses Using Computational Fluid Dynamics for Surgical Skill Training-A Retrospective Study.

Authors: Kiss-Pápai Levente, Reich Stefánia, Varga Júlia, Oosterlinck Wouter, Gloviczki Peter, Gasz Balázs

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Abstract

Background: Objective performance assessment is essential in surgical skill training, yet current methods are labor-intensive and focus on observing the trainee rather than the end-product of the procedure. Machine learning (ML) methods offer reproducible feedback but have mainly relied on kinematic or video data, often reducing assessment to binary or ternary classification. Our objective was to compare ML regression models predicting expert-assigned scores of vascular anastomoses from computational fluid dynamics (CFD) features of the final product. Additionally, we aimed to assess biomechanical plausibility of predictions. Methods: A total of 146 participants performed 419 end-to-side anastomoses on case-specific three-dimensional (3D) printed simulators. Anastomoses were digitized via 3D scanning, ranked by experts, and characterized using CFD-derived hemodynamic features. These served as input for linear models (Ridge, Partial Least Squares), support vector machines, and tree-based ensembles (Random Forest, Extremely Randomized Trees, and Extreme Gradient Boosting [XGBoost]), evaluated using 10-fold nested cross-validation with genetic hyperparameter optimization. Results: Inter-rater reliability of expert indicated strong agreement (intraclass correlation coefficient ICC3k = 0.846). XGBoost achieved the lowest mean root mean squared error of 0.758 (95% bootstrap CI: 0.722-0.799) and a coefficient of determination (R2) of 0.673 (0.617-0.725), with the most stable performance across folds. Shapley additive explanations (SHAP) identified the wall shear stress gradient, transverse wall shear stress, and maximum pressure as the most influential features-variables associated with intimal hyperplasia and atherosclerotic remodeling. Conclusions: Tree-based ensemble methods, particularly XGBoost, effectively modeled biomechanical properties against expert scores. Combining CFD and ML can provide reproducible, mechanistically relevant feedback in vascular surgical skill training.

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